DUST MANAGEMENT PLAN

For Lydiate Lane Quarry

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1.0 INTRODUCTION

Lydiate Lane Quarry is situated on Lydiate Lane, Farington, Leyland, approximately 1 mile north of Leyland town centre and immediately west of the West Coast Main Line railway. The site, which extracts and processes sand and gravel and accepts imported inert materials for landfill recovery, is located at National Grid Reference SD 552 238 (See Figure 1).

Figure 1. Location map for Lydiate Lane Quarry

1.1 Site Description

The Quarry material is mainly sand and gravel with soil and clay overburden. Lydiate Lane Quarry is operated by J A Jackson Contractors (Leyland) Limited and we supply to both companies and individuals locally across Lancashire and the wider North-West. The site also accepts imported inert waste materials for landfill recovery and progressive restoration of the worked-out void.

The Quarry extraction of sand and gravel and recovery of inert materials: Extraction, Stockpiling, Screening, Washing and Grading processes to produce the following products:

Washed building sand
Concreting sand
Graded gravel 4/10, 10/20, 20/40
Recycled inert aggregates
Inert tipping / void restoration

1.2 Aims

The aim of the Dust Management Plan is to:

a) Minimise dust generation and migration from the site.
b) Ensure nuisance caused to nearby receptors from dust is kept to a minimum.
c) To develop a dust minimisation strategy that shall be implemented by the site management; and
d) Ensure that operations at the site have consideration for potential dust generation.
e) Inform continuing improvements to dust/ particulate control and site management at the site and update the Dust Management Plan detailing such improvements.

This plan is integral to the Company EMS, Accident Management plan and Safe Operation Procedures.

Table 1. Potential receptors at Lydiate Lane Quarry

Ref	Name	Direction	Distance (m)
01	Oakfield	NNW	295
02	Bottoms Farm	WSW	324
03	Clock House Farm	E	695
04	M6 Motorway	ENE	494
05	Residential — Lydiate Lane (south)	SW	252
06	Clayton Farm	W	197

See Appendix B for plan

2.0 DUST CONTROL MEASURES

The following section outlines the control measures that will be undertaken on site to mitigate dust emissions from the identified sources of generation.

2.1 Dust Generating Activities

Potential dust emissions (not exhaustive) from the site may be generated from activities associated with:

- Vehicle movements in / around /out of the site. Tyres and exhausts may cause dust – keep vehicle speeds down, keep roads clear of mud, keep roads damp during dusty conditions.
- Loading and tipping operations – During this process dust may be given off through impact, therefore it is encouraged for tipping heights to be kept to a minimum. Mobile apps may be used to also suppress dust generation.
- Processing (Screening, Washing and Grading) of sand and gravel. Dust may arise due to impact (material against material, product coming off discharge belts). Material is dampened during washing prior to product being placed on the screening conveyor. Discharge belts to be kept to a minimum height.
- Handling and movement of stockpiles. Vehicles moving stockpiles will not over fill buckets / body of dumper, tipping heights will be kept low when emptying bucket.
- Wind blowing across stockpiled materials (washed sand, graded gravel, sub-base, recycled inert aggregates). In some conditions it may be necessary to cover the stockpile with plastic sheeting should the material be stored for a long period.
- Road sweeping may generate dust though the exhaust and brushes, this can be controlled by sufficient water being deployed from the spray bars.

In addition to the above, dust generation may be significant during periods of strong winds and dry weather.

Dust generation can also be attributed to other external influences.

- Quarry - sand and gravel extraction
- Quarry - Screening, washing and grading of aggregates
- Inert tipping operations within the void
- Road vehicles on Lydiate Lane and local highway network

2.2 Means of Prevention

In order to minimise potential generation of dust from the site, the following preventative control measures using best practicable means, shall be implemented by the site manager for the separately identified potential dust generating activities.

2.2.1 Vehicle Movements In/Out of Site

A) All haul and access roads within the site and at the site entrance shall be kept free from mud and debris at all times by manual clearing (Brooms, spades) and road sweeping. Mud and debris on access and haul roads shall be monitored daily by the site manager and cleaned when required. If this proves to be insufficient, a road sweeper will need to be provided. The public highway is also monitored for debris leaving the site.

B) The site management shall ensure adequate measures are used throughout the site to dampen surfaces (application of water through hoses / bowser / mobile apps) during periods of dry weather.

C) All vehicles and plant will be checked by the driver / operator to ensure that deposits of mud are not carried outside the site (signs of this will be visible on-site roads). Should mud become an issue then the installation of a wheel wash will be considered.

D) A site speed limit of 5 mph will be enforced for all vehicles to minimise the potential entrainment of dust into the atmosphere. All site roads are concrete.

2.2.2 Loading and Tipping Operations

E) All wastes handled on site shall be done so in a controlled manner, with consideration given to the potential for dust generation at all times.

F) Loading and tipping heights will be minimised to avoid uncontrolled dust emissions.

G) All vehicles will be sheeted when entering and leaving the site.

2.2.3 Handling and Movement of Stockpiles

H) The site manager will consider weather conditions at the site on a daily basis and shall have regard for high wind speeds. Wind speed and direction shall be measured on the site.

I) Where wind speeds are measured which are considered excessive (where visible dust clouds can be seen moving to the perimeter of the site), the site manager shall ensure that the movement of materials on site is controlled (reduced speeds / stopped/ addition suppression applied) until wind speeds reduce significantly.

J) The site management shall ensure appropriate measures are used throughout the site to dampen surfaces (application of water through hoses / bowser / mobile apps) during periods of dry weather. Such surfaces shall include stockpiles where appropriate.

2.2.4 Wind Blowing Across Stockpiles

K) Where necessary and during periods of dry conditions, water will be deployed to dampen material (use of binders will also be considered, along with sheeting) during stockpiling and made available as per J).

L) Disturbance of the surface of the stockpiles will be minimised to maintain an intact surface crust. Some stockpiles are in concrete block bays which act as windbreaks, stock pile heights are governed by planning.

2.2.5 Screening of Wastes

M) All inert handling/loading/screening operations on site shall be monitored by the site management, and if necessary appropriate measures shall be implemented to prevent dust generation.

N) Where dust suppression systems are incorporated into plant/machinery, they should be used to minimise dust generation where appropriate and maintained in workable condition at all times.

O) Operations around the operational machinery will be carried out in a controlled manner to prevent fall out of dust (Sprays around hoppers, nose bags on end of conveyor).

P) Screening operations will take place within the designated area and materials wetted prior to activities that could lead to dust generation where necessary.

2.2.6 Site Management

Q) The site manager shall ensure that a visual inspection of the activities is carried out at regular intervals during operational hours to assess the extent of dust being generated. In circumstances where visual dust inspection identifies a significant dust source, the site manager shall adopt appropriate dust suppression measures to prevent or minimise the dust being generated.

R) Dust suppression systems (mobile apps, plant suppression, Bowser dampening roads) and equipment used on site shall be maintained in good working order at all times.

S) Maintenance or repairs of dust suppression equipment and road / yard surfaces shall be carried out as soon as reasonably practicable and recorded within the relevant maintenance log.

T) Site operating personnel, including plant operators, will be supplied with dust masks, whenever necessary, and all plant cabs shall be maintained such that as far as reasonably practical the ingress of dust is minimised.

3.0 MANAGEMENT AND REVIEW PROCEDURE

The site manager shall be responsible for the control and management of dust at the site. Site management shall ensure that all personnel operating on site are adequately trained to implement the dust control measures.

If the control measures stated are implemented at the site, then dust generation should be kept to a minimum.

In the event that dust nuisance is caused to a nearby sensitive receptor, and a complaint is received by the site management, the 'Dust Action Plan' will be implemented.

4.0 DUST ACTION PLAN

If an activity at the site results in unacceptable levels of dust being generated, then that activity shall be closely monitored until sufficient measures using best practicable means are adopted which prevent or minimises the dust nuisance. The implementation of such measures will be the responsibility of the site manager. Conditions which may require the use of dust suppression at the site include the following:

- Dry surfaces where mud or debris is present.
- Any part of the site where movement of vehicles generate dust.
- Any part of the site where dust may be generated by wind.
- Stockpiles without cover.
- Screening activities – in the hopper, end of conveyors.
- Material handling operations; and
- Any other site activity which results in dust generation.

The site manager will be responsible for monitoring dust levels associated with the conditions and activities identified above. The site manager shall implement adequate dust suppression measures to control dust from any activity which causes dust nuisance.

4.1 Dust Monitoring

Upon receipt of a dust complaint the site manager shall be immediately notified. The site manager will record the details of the complaint on a Dust Complaints Form (see Annex A). This will trigger a levelled response.

- Level 1 - instigate investigation and possible monitoring dependent upon findings.
- Level 2 - improvement plan, include but not limited to improving working methods, mitigation measures, equipment and dust suppression methods.
- Level 3 – Re-Evaluate if the improvement plan has worked, including effects of weather conditions and infrastructure.
- Level 4 – Compliance confirmed.

4.2 Review of Techniques

In circumstances where the complaint is related to a previous period of working, the site manager shall meet with the Senior Manager, Health Safety and Environment Manager and site staff to establish the activities carried out during the previous working period. The activity identified as the potential dust source will then be monitored by the site manager. The manager will ensure that a review of dust suppression measures is carried out for any activity suspected of causing a dust complaint. Dust suppression measures will be adopted for any activity suspected as the cause of a dust complaint.

Where the site walkover by the site manager is able to identify the dust nuisance which caused complaint, appropriate dust suppression measures are to be adopted. The details of the dust source and the control measures adopted shall be recorded on the Dust Complaints Form.

4.3 Monitoring Points

If the dust source that led to a complaint cannot be established the site manager will arrange for dust monitoring to be undertaken (making reference to Control and Monitoring emissions for your environmental permit Nov 2018).

The dust monitoring point will be monitored for a full working day period (or longer, if required) and will comprise of a suitable monitoring system. The dust monitoring procedure for the monitoring system shall be followed in accordance with the manufacturer and best practice guidelines.

The dust monitoring sample will be collected after the monitoring period and sent for independent analysis. The dust monitoring results and findings shall be reported to the senior manager.

Details of dust control and mitigation measures will be reported by the senior manager to the EA. The nature of the complaint, the findings of any investigation, and mitigation measures adopted, will be recorded on the Dust Complaint Form, and then signed by the site manager. If the location of the complaint is known, we may wish to inform them of our findings and explain what we are doing on site. In the past the Company has engaged with the parish council and residents, however this has not been requested to continue but we will re- instigate if the need arises.

5.0 CONCLUSION

The extraction, processing and inert tipping activities on site should be a minimal issue with regards to dust as control measures will be used. The site is well screened and inert tipping takes place within the worked-out void which provides natural shelter from wind.

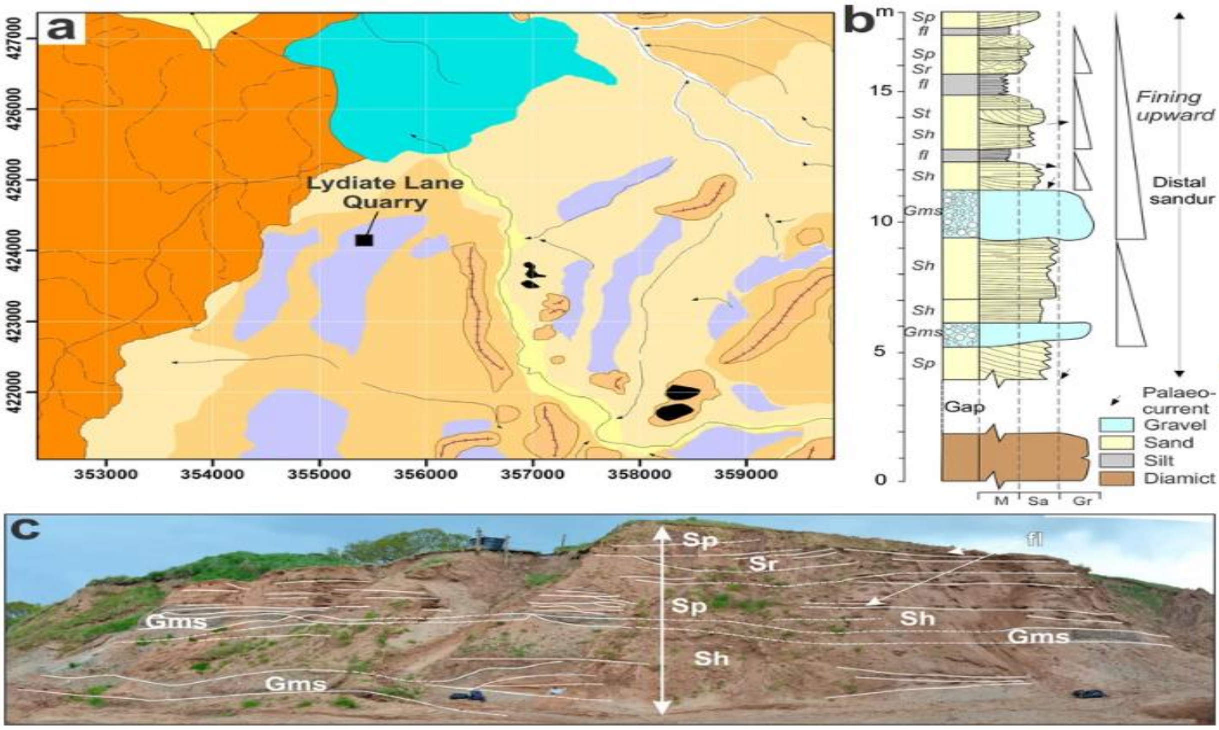
If the control measures stated are implemented at the site, it is considered that dust generation should be kept to a minimum and that nuisance to nearby receptors should be avoided in normal circumstances.

In the event that dust nuisance is caused to a nearby sensitive receptor, and a complaint is received regarding dust migrating from the site, the dust action plan shall be implemented. This may require a period of monitoring to be implemented to inform operations and establish dust emissions levels with targets for improvements.

APPENDIX A - DUST COMPLAINTS FORM

DUST COMPLAINTS FORM		
Complaint Received From:	**Date of Event**	
	Time of Event	
	Direction from Site	
Investigation		
Wind Direction During Event	**Complaint Substantiated**	
	YES	NO
Activities at Time of Event	**Excessive Dust Emissions Identified**	
	YES	NO
Actions/Mitigation Measures		
Measures Implemented Date		
Feedback to EA (Date)		
Investigation Complete (Date)		
Review		
Signed Off **(Site Manager)**		

APPENDIX B – SITE PLANS



(a) Geomorphic map of the area around Lydiate Lane Quarry (symbols as in Fig. 1). (b) Summary stratigraphy of the sequence at Lydiate Lane Quarry. (c) Photograph of the sequence identifying key lithofacies. The main unit boundaries are shown by the white lines. Log location is indicated by the white arrow.